

autonomous agent evaluation. The evaluation of the agents play significant roles in this domain. However, both subjective and objective evaluation have their own strengths and weakness. Maybe, in practice, they should be combined to comprehensively evaluate the agents. We summarize the correspondence between the previous work and these evaluation strategies in Table 3.

5 Related Surveys

With the vigorous development of large language models, a variety of comprehensive surveys have emerged, providing detailed insights into various aspects. [175] extensively introduces the background, main findings, and mainstream technologies of LLMs, encompassing a vast array of existing works. On the other hand, [176] primarily focus on the applications of LLMs in various downstream tasks and the challenges associated with their deployment. Aligning LLMs with human intelligence is an active area of research to address concerns such as biases and illusions. [177] have compiled existing techniques for human alignment, including data collection and model training methodologies. Reasoning is a crucial aspect of intelligence, influencing decision-making, problem-solving, and other cognitive abilities. [178] presents the current state of research on LLMs’ reasoning abilities, exploring approaches to improve and evaluate their reasoning skills. [179] propose that language models can be enhanced with reasoning capabilities and the ability to utilize tools, termed Augmented Language Models (ALMs). They conduct a comprehensive review of the latest advancements in ALMs. As the utilization of large-scale models becomes more prevalent, evaluating their performance is increasingly critical. [180] shed light on evaluating LLMs, addressing what to evaluate, where

to evaluate, and how to assess their performance in downstream tasks and societal impact. [181] also discusses the capabilities and limitations of LLMs in various downstream tasks. The aforementioned research encompasses various aspects of large models, including training, application, and evaluation. However, prior to this paper, no work has specifically focused on the rapidly emerging and highly promising field of LLM-based Agents. In this study, we have compiled 100 relevant works on LLM-based Agents, covering their construction, applications, and evaluation processes.

6 Challenges

While previous work on LLM-based autonomous agent has obtained many remarkable successes, this field is still at its initial stage, and there are several significant challenges that need to be addressed in its development. In the following, we present many representative challenges.

6.1 Role-playing Capability

Different from traditional LLMs, autonomous agent usually has to play as specific roles (*e.g.*, program coder, researcher and chemist) for accomplishing different tasks. Thus, the capability of the agent for role-playing is very important. Although LLMs can effectively simulate many common roles such as movie reviewers, there are still various roles and aspects that they struggle to capture accurately. To begin with, LLMs are usually trained based on web-corpus, thus for the roles which are seldom discussed on the web or the newly emerging roles, LLMs may not simulate them well. In addition, previous research [30] has shown that existing LLMs may not well model the human cognitive psychology characters, leading to the lack of self-awareness

in conversation scenarios. Potential solution to these problems may include fine-tuning LLMs or carefully designing the agent prompts/architectures [182]. For example, one can firstly collect real-human data for uncommon roles or psychology characters, and then leverage it to fine-tune LLMs. However, how to ensure that fine-tuned model still perform well for the common roles may pose further challenges. Beyond fine-tuning, one can also design tailored agent prompts/architectures to enhance the capability of LLM on role-playing. However, finding the optimal prompts/architectures is not easy, since their designing spaces are too large.

6.2 Generalized Human Alignment

Human alignment has been discussed a lot for traditional LLMs. In the field of LLM-based autonomous agent, especially when the agents are leveraged for simulation, we believe this concept should be discussed more in depth. In order to better serve human-beings, traditional LLMs are usually fine-tuned to be aligned with correct human values, for example, the agent should not plan to make a bomb for avenging society. However, when the agents are leveraged for real-world simulation, an ideal simulator should be able to honestly depict diverse human traits, including the ones with incorrect values. Actually, simulating the human negative aspects can be even more important, since an important goal of simulation is to discover and solve problems, and without negative aspects means no problem to be solved. For example, to simulate the real-world society, we may have to allow the agent to plan for making a bomb, and observe how it will act to implement the plan as well as the influence of its behaviors. Based on these observations, people can make better actions to stop similar behaviors in real-world society. Inspired by the above case, maybe

an important problem for agent-based simulation is how to conduct generalized human alignment, that is, for different purposes and applications, the agent should be able to align with diverse human values. However, existing powerful LLMs including ChatGPT and GPT-4 are mostly aligned with unified human values. Thus, an interesting direction is how to “realign” these models by designing proper prompting strategies.

6.3 Prompt Robustness

To ensure rational behavior in agents, it’s a common practice for designers to embed supplementary modules, such as memory and planning modules, into LLMs. However, the inclusion of these modules necessitates the development of more complex prompts in order to facilitate consistent operation and effective communication. Previous research [183, 184] has highlighted the lack of robustness in prompts for LLMs, as even minor alterations can yield substantially different outcomes. This issue becomes more pronounced when constructing autonomous agents, as they encompass not a single prompt but a prompt framework that considers all modules, wherein the prompt for one module has the potential to influence others. Moreover, the prompt frameworks can vary significantly across different LLMs. The development of a unified and resilient prompt framework applicable across diverse LLMs remains a critical and unresolved challenge. There are two potential solutions to the aforementioned problems: (1) manually crafting the essential prompt elements through trial and error, or (2) automatically generating prompts using GPT.

6.4 Hallucination

Hallucination poses a fundamental challenge for LLMs, characterized by the models’ tendency to

produce false information with a high level of confidence. This challenge is not limited to LLMs alone but is also a significant concern in the domain of autonomous agents. For instance, in [185], it was observed that when confronted with simplistic instructions during code generation tasks, the agent may exhibit hallucinatory behavior. Hallucination can lead to serious consequences such as incorrect or misleading code, security risks, and ethical issues [185]. To mitigate this issue, incorporating human correction feedback directly into the iterative process of human-agent interaction presents a viable approach [23]. More discussions on the hallucination problem can be seen in [175].

6.5 Knowledge Boundary

A pivotal application of LLM-based autonomous agents lies in simulating diverse real-world human behaviors [20]. The study of human simulation has a long history, and the recent surge in interest can be attributed to the remarkable advancements made by LLMs, which have demonstrated significant capabilities in simulating human behavior. However, it is important to recognize that the power of LLMs may not always be advantageous. Specifically, an ideal simulation should accurately replicate human knowledge. In this context, LLMs may display overwhelming capabilities, being trained on a vast corpus of web knowledge that far exceeds what an average individual might know. The immense capabilities of LLMs can significantly impact the effectiveness of simulations. For instance, when attempting to simulate user selection behaviors for various movies, it is crucial to ensure that LLMs assume a position of having no prior knowledge of these movies. However, there is a possibility that LLMs have already acquired information about these movies. Without implementing appropriate

strategies, LLMs may make decisions based on their extensive knowledge, even though real-world users would not have access to the contents of these movies beforehand. Based on the above example, we may conclude that for building believable agent simulation environment, an important problem is how to constrain the utilization of user-unknown knowledge of LLM.

6.6 Efficiency

Due to their autoregressive architecture, LLMs typically have slow inference speeds. However, the agent may need to query LLMs for each action multiple times, such as extracting information from memory, make plans before taking actions and so on. Consequently, the efficiency of agent actions is greatly affected by the speed of LLM inference.

7 Conclusion

In this survey, we systematically summarize existing research in the field of LLM-based autonomous agents. We present and review these studies from three aspects including the construction, application, and evaluation of the agents. For each of these aspects, we provide a detailed taxonomy to draw connections among the existing research, summarizing the major techniques and their development histories. In addition to reviewing the previous work, we also propose several challenges in this field, which are expected to guide potential future directions.

Acknowledgement

This work is supported in part by National Natural Science Foundation of China (No. 62102420), Beijing Outstanding Young Scientist Program NO. BJJWZYJH012019100020098, Intelligent Social

Governance Platform, Major Innovation & Planning Interdisciplinary Platform for the "Double-First Class" Initiative, Renmin University of China, Public Computing Cloud, Renmin University of China, fund for building world-class universities (disciplines) of Renmin University of China, Intelligent Social Governance Platform.

References

1. Mnih V, Kavukcuoglu K, Silver D, Rusu A A, Veness J, Bellemare M G, Graves A, Riedmiller M, Fiedel A K, Ostrovski G, others . Human-level control through deep reinforcement learning. *nature*, 2015, 518(7540): 529–533
2. Lillicrap T P, Hunt J J, Pritzel A, Heess N, Erez T, Tassa Y, Silver D, Wierstra D. Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*, 2015
3. Schulman J, Wolski F, Dhariwal P, Radford A, Klimov O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017
4. Haarnoja T, Zhou A, Abbeel P, Levine S. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In: *International conference on machine learning*. 2018, 1861–1870
5. Brown T, Mann B, Ryder N, Subbiah M, Kaplan J D, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A, others . Language models are few-shot learners. *Advances in neural information processing systems*, 2020, 33: 1877–1901
6. Radford A, Wu J, Child R, Luan D, Amodei D, Sutskever I, others . Language models are unsupervised multitask learners. *OpenAI blog*, 2019, 1(8): 9
7. Achiam J, Adler S, Agarwal S, Ahmad L, Akkaya I, Aleman F L, Almeida D, Altenschmidt J, Altman S, Anadkat S, others . Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023
8. Anthropic . Model card and evaluations for claude models. <https://www-files.anthropic.com/production/images/Model-Card-Claude-2.pdf?ref=maginataive.com>, 2023
9. Touvron H, Lavril T, Izacard G, Martinet X, Lachaux M A, Lacroix T, Rozière B, Goyal N, Hambro E, Azhar F, others . Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023
10. Touvron H, Martin L, Stone K, Albert P, Almahairi A, Babaei Y, Bashlykov N, Batra S, Bhargava P, Bhosale S, others . Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023
11. Chen X, Li S, Li H, Jiang S, Qi Y, Song L. Generative adversarial user model for reinforcement learning based recommendation system. In: *International Conference on Machine Learning*. 2019, 1052–1061
12. Shinn N, Cassano F, Gopinath A, Narasimhan K, Yao S. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 2024, 36
13. Shen Y, Song K, Tan X, Li D, Lu W, Zhuang Y. Hugginggpt: Solving ai tasks with chatgpt and its friends in hugging face. *Advances in Neural Information Processing Systems*, 2024, 36
14. Qin Y, Liang S, Ye Y, Zhu K, Yan L, Lu Y, Lin Y, Cong X, Tang X, Qian B, others . Toolllm: Facilitating large language models to master 16000+ real-world apis. *arXiv preprint arXiv:2307.16789*, 2023
15. Schick T, Dwivedi-Yu J, Dessì R, Raileanu R, Lomeli M, Hambro E, Zettlemoyer L, Cancedda N, Scialom T. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 2024, 36
16. Zhu X, Chen Y, Tian H, Tao C, Su W, Yang C, Huang G, Li B, Lu L, Wang X, others . Ghost in the minecraft: Generally capable agents for open-world environments via large language models with text-based knowledge and memory. *arXiv preprint arXiv:2305.17144*, 2023
17. Sclar M, Kumar S, West P, Suhr A, Choi Y, Tsvetkov Y. Minding language models' (lack of) theory of mind: A plug-and-play multi-character belief tracker. *arXiv preprint arXiv:2306.00924*, 2023
18. Qian C, Cong X, Yang C, Chen W, Su Y, Xu J, Liu Z, Sun M. Communicative agents for software development. *arXiv preprint arXiv:2307.07924*, 2023
19. al. e C. Agentverse. <https://github.com/OpenBMB/AgentVerse>, 2023
20. Park J S, O'Brien J, Cai C J, Morris M R, Liang P, Bernstein M S. Generative agents: Interactive simulators of human behavior. In: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 2023, 1–22
21. Wang L, Zhang J, Chen X, Lin Y, Song R, Zhao W X, Wen J R. Recagent: A novel simulation paradigm for recommender systems. *arXiv preprint arXiv:2306.02552*, 2023
22. Zhang H, Du W, Shan J, Zhou Q, Du Y, Tenenbaum J B, Shu T, Gan C. Building cooperative embodied agents modularly with large language models. *arXiv*

- preprint arXiv:2307.02485, 2023
23. Hong S, Zheng X, Chen J, Cheng Y, Wang J, Zhang C, Wang Z, Yau S K S, Lin Z, Zhou L, others . Metagpt: Meta programming for multi-agent collaborative framework. arXiv preprint arXiv:2308.00352, 2023
 24. Dong Y, Jiang X, Jin Z, Li G. Self-collaboration code generation via chatgpt. arXiv preprint arXiv:2304.07590, 2023
 25. Safdari M, Serapio-García G, Crepy C, Fitz S, Romero P, Sun L, Abdulhai M, Faust A, Matarić M. Personality traits in large language models. arXiv preprint arXiv:2307.00184, 2023
 26. Johnson J A. Measuring thirty facets of the five factor model with a 120-item public domain inventory: Development of the ipip-neo-120. *Journal of research in personality*, 2014, 51: 78–89
 27. John O P, Donahue E M, Kentle R L. Big five inventory. *Journal of Personality and Social Psychology*, 1991
 28. Deshpande A, Murahari V, Rajpurohit T, Kalyan A, Narasimhan K. Toxicity in chatgpt: Analyzing persona-assigned language models. arXiv preprint arXiv:2304.05335, 2023
 29. Argyle L P, Busby E C, Fulda N, Gubler J R, Rytting C, Wingate D. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 2023, 31(3): 337–351
 30. Fischer K A. Reflective linguistic programming (rlp): A stepping stone in socially-aware agi (socialagi). arXiv preprint arXiv:2305.12647, 2023
 31. Rana K, Haviland J, Garg S, Abou-Chakra J, Reid I, Suenderhauf N. Sayplan: Grounding large language models using 3d scene graphs for scalable robot task planning. In: 7th Annual Conference on Robot Learning. 2023
 32. Zhu A, Martin L, Head A, Callison-Burch C. Calypso: Llms as dungeon master’s assistants. In: Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment. 2023, 380–390
 33. Wang Z, Cai S, Chen G, Liu A, Ma X, Liang Y. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents. arXiv preprint arXiv:2302.01560, 2023
 34. Lin J, Zhao H, Zhang A, Wu Y, Ping H, Chen Q. Agentsims: An open-source sandbox for large language model evaluation. arXiv preprint arXiv:2308.04026, 2023
 35. Liang X, Wang B, Huang H, Wu S, Wu P, Lu L, Ma Z, Li Z. Unleashing infinite-length input capacity for large-scale language models with self-controlled memory system. arXiv preprint arXiv:2304.13343, 2023
 36. Ng Y, Miyashita D, Hoshi Y, Morioka Y, Torii O, Kodama T, Deguchi J. Simplyretrieve: A private and lightweight retrieval-centric generative ai tool. arXiv preprint arXiv:2308.03983, 2023
 37. Huang Z, Gutierrez S, Kamana H, MacNeil S. Memory sandbox: Transparent and interactive memory management for conversational agents. In: Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology. 2023, 1–3
 38. Wang G, Xie Y, Jiang Y, Mandlekar A, Xiao C, Zhu Y, Fan L, Anandkumar A. Voyager: An open-ended embodied agent with large language models. arXiv preprint arXiv:2305.16291, 2023
 39. Zhong W, Guo L, Gao Q, Wang Y. Memorybank: Enhancing large language models with long-term memory. arXiv preprint arXiv:2305.10250, 2023
 40. Hu C, Fu J, Du C, Luo S, Zhao J, Zhao H. Chatdb: Augmenting llms with databases as their symbolic memory. arXiv preprint arXiv:2306.03901, 2023
 41. Modarressi A, Imani A, Fayyaz M, Schütze H. Ret-llm: Towards a general read-write memory for large language models. arXiv preprint arXiv:2305.14322, 2023
 42. Schuurmans D. Memory augmented large language models are computationally universal. arXiv preprint arXiv:2301.04589, 2023
 43. Zhao A, Huang D, Xu Q, Lin M, Liu Y J, Huang G. Expel: Llm agents are experiential learners. arXiv preprint arXiv:2308.10144, 2023
 44. Huang W, Abbeel P, Pathak D, Mordatch I. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In: International Conference on Machine Learning. 2022, 9118–9147
 45. Wei J, Wang X, Schuurmans D, Bosma M, Xia F, Chi E, Le Q V, Zhou D, others . Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 2022, 35: 24824–24837
 46. Kojima T, Gu S S, Reid M, Matsuo Y, Iwasawa Y. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 2022, 35: 22199–22213
 47. Raman S S, Cohen V, Rosen E, Idrees I, Paulius D, Tellex S. Planning with large language models via corrective re-prompting. In: NeurIPS 2022 Foundation Models for Decision Making Workshop. 2022
 48. Xu B, Peng Z, Lei B, Mukherjee S, Liu Y, Xu D. Rewoo: Decoupling reasoning from observations for efficient augmented language models. arXiv preprint arXiv:2305.18323, 2023
 49. Lin B Y, Fu Y, Yang K, Brahman F, Huang S, Bhagavatula C, Ammanabrolu P, Choi Y, Ren X. Swiftsage: